



Society of Petroleum Engineers

SPE-229103-MS

Facilitating the Future: Industrial Case Study of a Facilitated Transport Membrane Pilot for PostCombustion CO₂ Capture

C. Krenn and A. Meißl, OMV Exploration & Production GmbH, Austria; B. Burghardt, M. Roberts, and C. Parrish, Ardent Process Technologies, United States of America

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229103-MS](https://doi.org/10.2118/229103-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

As global efforts to mitigate climate change intensify, carbon capture (CC) technologies are becoming increasingly vital in reducing greenhouse gas emissions from industrial and energy sectors, making them necessary to reach industrial net zero goal pledges([International Energy Agency 2020](#)). Among these CC-technologies, membrane-based carbon capture has emerged as a promising alternative to conventional and well-known solvent-based systems (like amines) due to its lower energy requirements, overall smaller footprint, and operational simplicity ([Kaushik 2024](#)).

Facilitated transport membranes (FTMs) offer distinct advantages over conventional polymeric membranes, including enhanced selectivity and permeability for CO₂ due to the use of reversible chemical carriers. These carriers enable selective and rapid CO₂ transport even at low partial pressures, making FTMs particularly attractive for post-combustion capture applications where conventional membrane economics struggle due to significantly higher compression costs. To be effective in the field, an FTM would need to also be chemically resistant to withstand process conditions, and a humidification method must be used to ensure efficiency of the carrier mechanism.

This paper presents the design, deployment, operational insights and results from a pilot system utilizing FTMs for CO₂ separation from actual flue gases. The pilot was fully integrated, treating flue gas from an incinerator unit located in a sour gas treatment plant in the Vienna Basin.

Due to the integration into a live plant and the treatment of actual flue gas this pilot allows an evaluation of membrane performance under realistic process conditions, addressing key parameters such as CO₂ capture rate, membrane stability, and process integration challenges. By translating pilot-scale advantages into a practical industrial setting, this work demonstrates the potential of FTMs to deliver lower energy penalties and improved capture efficiency, contributing to the development of next-generation CC-technologies.

In this paper we will discuss the high separation in the field (over 40 mol.% enrichment of the CO₂ from feed to permeate achieved) as well as the performance behavior in a wide variety of operating conditions. After nearly 1,000 hours in the field, the membrane matched expectations set by lab testing when a novel humidification method was employed. We will discuss the effects of product design, such as hollow fiber

module design, the effect of pressure drops, and overall system design to optimally run under very low driving forces. The design resulting from this technology can reduce the system complexity by removing two unit operations and reducing area and compression needs by 50% compared to traditional membranes and offers a 70% reduction in cost of capture with 80% energy reduction compared to state of the art amines.

The FTM technology deployed in this application utilizes a proprietary advanced composite membrane in a hollow fiber configuration. A novel humidification system ensures high FTM performance at commercially-relevant conditions.

Introduction

As the need to mitigate climate change by capturing CO₂ rises so do the possible challenges that come with it. One of the industries that is seeking to find efficient and cost effective ways of doing so is the oil and gas industry which is responsible for around 5.1 billion tons of greenhouse gas emissions per year ([International Energy Agency 2023](#)). Oil and gas fields in the Vienna basin were first explored in the first half of the 20th century and have been in production for more than 70 years. All the produced sour gas streams of the Vienna basin are bundled and treated in the sour gas station Aderklaa where most of the natural gas's CO₂ is being removed via amine washers. Following this step the concentrated CO₂ stream is mixed with steam boiler flue gases, off gas from the onsite Claus-process as well as multiple other process gas streams and thermally treated by an incinerator before being released into the environment. The diversity of gas streams feeding this incinerator and therefore the high variation in process conditions as well as the levels of contaminants made it the preferred candidate to field test the robustness of the new FTM technology.

The general idea behind choosing FTMs as a possible decarbonization technology was that they offer solutions to some of the challenges described below, related to the integration of CC-technologies into brownfield projects or plants just like the one in Aderklaa. While implementing new technologies into already existing infrastructure can be advantageous it is also true that such projects are often limited by many factors such as CAPEX constraints due to limited lifetime of host-plants, footprint constraints and the need to reduce energy costs as much as possible all while reaching the targeted CO₂ specifications for transport, storage or utilization.

The evaluation of the suitability of carbon capture membrane technology for deployment in brownfield applications utilized a fully integrated pilot test stand treating flue gas from an incinerator unit located on-site. This field test had the goal to test the robustness of the tested FTMs as well as gain valuable field data showing the efficiency and flexibility of the technology. The technology evaluation was conducted in two phases. The first phase evaluated the performance of the membrane at operating conditions representative of membrane use cases in complete systems designed to deliver CO₂ at pipeline specifications. The second phase evaluated the stability of the FTM technology while enriching carbon from the untreated flue gas stream. By translating pilot-scale advantages into a practical industrial setting, the pilot campaign demonstrates the potential of FTMs to deliver lower energy penalties and improved capture efficiency, contributing to the development of next-generation CC-technologies. Derived from the field data in combination with existing lab tests, estimations of future footprints and capture costs are made.

Key Performance Indicators (KPIs) for the Membrane System

Models driven from pilot data are utilized for system designs optimizing levelized carbon capture cost and energy usage. In this way, the membrane technology is compared to current commercial carbon capture systems using amine absorption/desorption as the main capture technology. [Table 1](#) shows a comparison between costs and energy of capture for amine-based systems, targets from piloting, and results from the pilot unit. The cost estimates for membrane systems incorporate electricity costs at \$60/MWh, maintenance costs, installation factors, and cost of capital for membrane systems operating at 90% CO₂ recovery. The

amine system data is reported from an engineering design report for an amine system operating at 90% CO₂ recovery (Shah 2022).

Table 1—Carbon Capture Membrane System Pilot KPIs

	Amine Systems	Pilot Target	Pilot Result	Improved design based on pilot learnings**
Cost of Capture [\$/ton CO ₂]	> \$71	\$46	\$50-65	\$28-43
Heat* [kWh/ton CO ₂]	720	—	—	—
Electricity [kWh/ton CO ₂]	135	302	256	154

*

Heat supplied from low pressure steam.

**

Simulation and pilot results. Cost savings resulting from process and membrane optimization based on pilot unit data.

The KPI's for the membrane pilot unit then are evaluated via the resulting cost of capture and energy of capture in comparison to the targets for the pilot project and to the commercially available amine systems. In comparison to the previous target of \$46 /ton of carbon captured, the results from the pilot unit yielded \$50–65 /ton, i.e. within 10% of the target value. The results from the pilot also yielded electrical energy usage of 256 kWh/ton, less than the target energy usage of 302 kWh/ton, and less than the commercial amine system energy usage of greater than 720 kWh/ton of heat supplied by low pressure steam and 135 kWh/ton of electrical energy. Modelled improvements to the system design enable further energy and cost of capture reductions as the membrane technology progresses toward commercialization and a lower capture cost of \$28–43/ton. Manufacturing improvements leading to increased performance could enable the same low cost in industries with low starting concentrations (i.e. FCC, boilers).

Many reports assess membrane performance on the basis of permeance and selectivity alone (Robert F Dejacco 2025). To ensure systems are designed to meet real-world conditions, this analysis was optimized for levelized cost of carbon (LCOC) and energy. The permeance is derived from Fickian transport assuming activity coefficients are close to one, that the diffusivity of the permeating species is constant, that the concentrations in the fluid phase are in equilibrium with the membrane governed by Henry's law, and that the membrane is isothermal with the surrounding fluid phases (Joseph D. Henry Jr. 1997). Using these assumptions the permeance is defined locally as the ratio of flux of a given species per the partial pressure difference of that species across the membrane (Joseph D. Henry Jr. 1997). The selectivity is then defined as the ratio of permeances of the higher permeance to lower permeance species. In the case of carbon capture membranes, it's desired that the CO₂ permeance is high and the N₂ permeance is low, such that the CO₂/N₂ selectivity is high.

Evaluating membranes for suitability in a commercial application on the basis of permeance and selectivity is problematic. The assumptions defining permeance ignore industrially relevant phenomena such as non-dilute permeation in the active layer and temperature variation from the Joule-Thompson effect. Since permeance is a local property, a model representative of the geometry and local concentrations inside the membrane must be used to estimate the permeance. While definitions of a global module permeance exist (Runhong Qi 1996), these estimations are not effective for predicting module behavior in a commercial system. Since membrane properties are inadequate for comparison to current commercial solutions, suitability of the technology is best assessed through the resulting levelized cost of capture and energy usage for system designs utilizing membranes for carbon capture.

Technology Overview

Facilitated transport membranes (FTMs) offer distinct advantages over conventional polymeric membranes, including enhanced selectivity and permeability for CO₂ due to the use of reversible chemical carriers. These carriers enable selective and rapid CO₂ transport even at low partial pressures, making FTMs particularly attractive for post-combustion capture applications where conventional membrane economics struggle due to significantly higher compression costs. To be effective in the field, the FTM needs to be chemically resistant to withstand process conditions and needs a humidification method to ensure efficiency of the carrier mechanism. The membranes tested in this pilot unit system were developed for this application to be resistant to degradation in the presence of acid gases and their oxidative degradation products and provide favorable CO₂ enrichment and flux for use in a commercial application.

The enrichment of CO₂ in the permeate stream is driven by differences in permeation rate, or flux, across the membrane. The flux for each species (i) is defined as the molar flowrate of each species (N_{Pi}) across the membrane per the membrane area (A), where the molar flowrate of a species (N_{Pi}) is the product of the total permeate molar flowrate (N_P) and the permeate mole fraction of that species (y_{Pi}). As CO₂ permeates the membrane at a higher rate than the N₂ and O₂ present in the feed stream, the gas retained on the feed side of the membrane is depleted in CO₂ and the gas permeating the membrane is enriched in CO₂.

The performance of the membrane was evaluated via the enrichment of CO₂ in the permeate (y_{P,CO_2}) relative to that of the feed stream (y_{F,CO_2}) (Equation 1) and depletion of CO₂ in the retentate stream (y_{R,CO_2}) relative to the feed stream (y_{F,CO_2}) (Equation 2) as well as the flux of CO₂ across the membrane. The stability was evaluated based on the stability of the CO₂ enrichment and Flux. The CO₂ Flux (J_{CO_2}) is defined as the product of the molar permeate flowrate (N_P) and the permeate CO₂ mole fraction (y_{P,CO_2}) divided by the membrane area (A) as shown in Equation 3.

$$\text{Enrichment in Permeate} = y_{P,CO_2} - y_{F,CO_2} \quad (1)$$

$$\text{Depletion in Retentate} = y_{R,CO_2} - y_{F,CO_2} \quad (2)$$

$$J_{CO_2} = \frac{N_P y_{P,CO_2}}{A} \quad (3)$$

In addition to flux and purity, a third metric is relevant for evaluating membranes at scale. The stage cut is defined as the ratio of permeate to feed flow (Equation 4). In a commercial system, stage cut tunes CO₂ purity and recovery leaving a single stage. As stage cut increases, CO₂ recovery and retentate N₂ purity increase at the expense of CO₂ purity in the permeate. Understanding membrane performance across the commercially relevant stage cut window is critical for sizing a system to target certain product specifications.

$$\text{Stage Cut} = \frac{N_P}{N_F} \quad (4)$$

Process Overview

Due to the integration into a plant and the treatment of actual flue gas, this pilot allows an evaluation of membrane performance under realistic process conditions, addressing key parameters such as CO₂ capture rate, membrane stability, and process integration challenges. The pilot system is designed to operate for the duration of short test campaigns at commercially relevant operating conditions and to operate continuously for extended runs.

The system drew gas from the flue source. Gas was then compressed and fed to the membrane element contained in the membrane housing. The compression of the feed gas provides the pressure differential across the membrane required for permeation. The membrane and process streams were heated, and vacuum was pulled on the permeate side of the membrane in order to increase permeation rate and permeate CO₂

purity. The pressure achieved by the vacuum is reproducible at industrial scale and is intended to be representative of the membrane's use in a commercial system.

Figure 1 shows the configuration of process flows around the membrane as well as instrumentation and equipment used to control operating conditions. A needle valve (NV) was used to set the feed flowrate, which was measured with a Bronkhorst Thermal Mass Flow Meter (T-MFM). Pressure transducers (P) were used to measure the feed and permeate pressures, and embedded J-Type thermocouples (T) were used to measure the feed, permeate, and retentate temperatures. There is little pressure drop from the feed to retentate, and the feed/retentate pressures are set with a backpressure regulator. The retentate flowrate is measured with a Bronkhorst Coriolis mass flowmeter (C-MFM) and the permeate flowrate was measured with a Brooks thermal mass flow meter (T-MFM). The permeate pressure was controlled via a spillback line from the vacuum pump discharge to suction using a needle valve. Live monitoring of pressure transducers, flowmeters, thermocouples, and pressure transducers was used to collect data for analysis. The mole fractions of CO₂ and O₂ were measured using a calibrated Quantek 902P analyzer, which utilizes an IR cell to determine CO₂ mole fraction and an electrochemical cell to determine O₂ mole fraction. Streams were sampled in a sequence over a set cycle time.

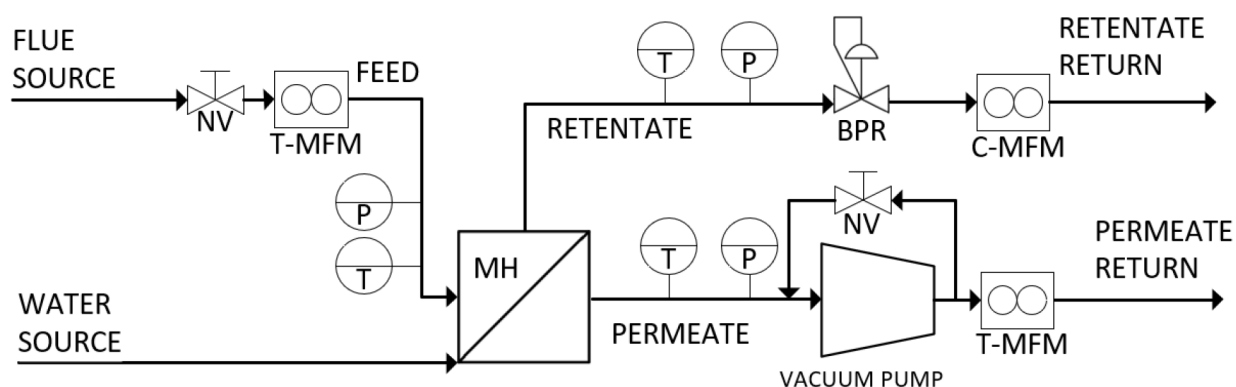


Figure 1—Process Flow Diagram of Membrane Pilot Unit. Detailed information regarding process equipment is described in the Process Overview text.

Results

In the following chapters the actual field data of the pilot is benchmarked against its original laboratory test data. Laboratory data is always ideal, the goal is to decrease the delta as much as possible.

Verification of accuracy of data collected

To evaluate the performance of the membrane at commercially relevant conditions, the enrichment of CO₂ in the permeate and depletion of CO₂ in the retentate were monitored over time alongside the CO₂ flux. The stability of the membrane was evaluated at a commercially relevant condition over 430 hours of operation on real flue gas.

To verify the accuracy of the data collected, the major component CO₂ mass balance error was computed from the flow measurements around the membrane. The relative CO₂ mass balance error is the CO₂ mass balance error relative to incoming flow. The mass balance error was computed such that a negative mass balance error indicates either a loss of mass between entering and exiting flowmeters or error in measurement appearing to be a loss of mass. These definitions were used for relative CO₂ mass balance error (equation 5) and relative CO₂ mass balance error excluding the feed stream (equation 6):

$$\text{Relative CO}_2 \text{ mass balance error} = \frac{(M_{PXP,CO_2} + M_{RXR,CO_2}) - x_{F,CO_2}}{M_{F XF,CO_2}} \quad (5)$$

$$\text{Relative CO}_2 \text{ mass balance error (excl. feed)} = \frac{(M_P x_{P,CO_2} + M_R x_{R,CO_2}) - (M_P + M_R) x_{F,CO_2}}{(M_P + M_R) x_{F,CO_2}} \quad (6)$$

Figure 2 shows that the relative CO₂ mass balance error is quite high (typically > 20% loss of mass). To determine the source of the high mass balance error, different methods were used to remove measurements from the calculation. A significant improvement in mass balance error when removing the flow measurement from a certain stream reveals the cause of the mass balance error. By removing the feed flowrate from the mass balance error estimation, the mass balance error is considerably improved from -20% to ±3%. This improvement demonstrates that either a loss of mass between the feed flowmeter and the membrane or an error in the feed flow measurement is responsible for the mass balance error; to ensure accuracy of data interpretation, this feed flowrate was not used for analysis of membrane performance (i.e. the feed flow was calculated from balance closure via the retentate and permeate flowmeters).

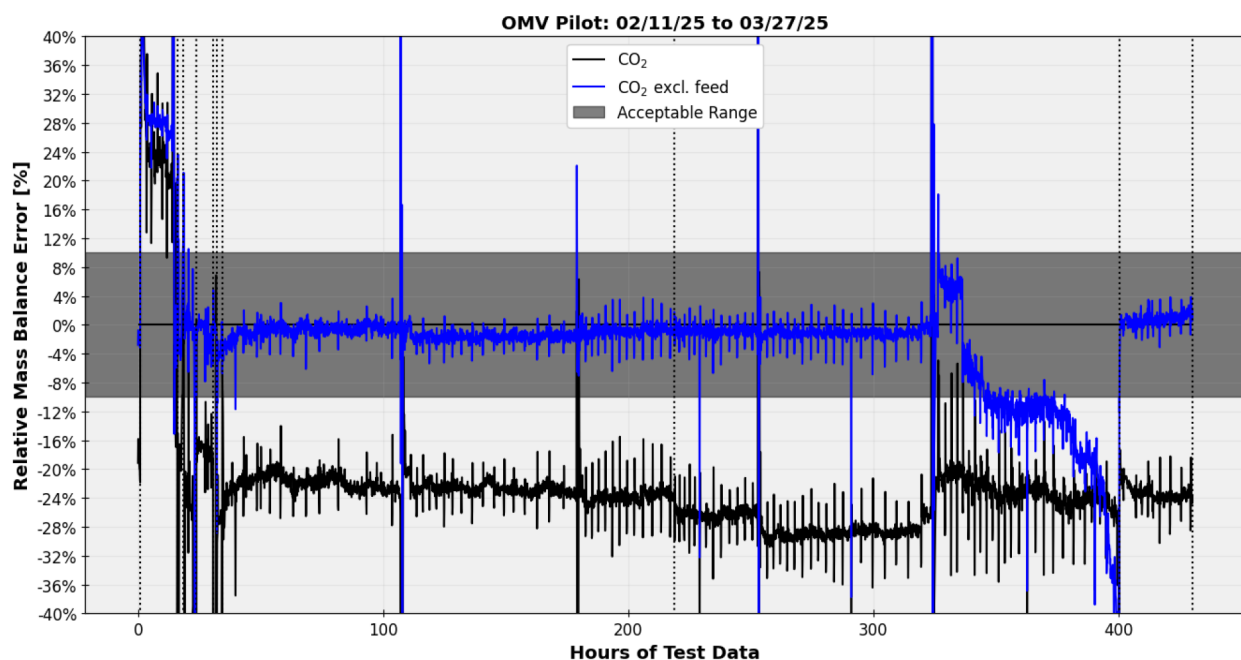


Figure 2—Relative CO₂ mass balance error and CO₂ mass balance error excluding the feed flowrate.

The significant improvement in mass balance error when the feed flowrate is removed from the calculation suggests that the feed flowrate is the major source of mass balance error. This indicates either a loss of mass between the feed flowmeter and the membrane or an incorrect reading at the feed flowmeter. For this reason, the feed flowrate was not used for analysis of membrane performance.

Testing Phase 1: Comparison to expectations and operating condition trends

The first phase evaluated the performance of the membrane at commercially applicable operating conditions in comparison to expected results. Figure 3 shows the permeate, feed, and retentate CO₂ mole fractions as well as CO₂ flux in comparison to a band of expectations determined by the expected membrane performance. Please note that flux is reported in arbitrary units on an absolute scale to protect proprietary information. These results confirmed that the membrane operating in the field is behaving as expected; the agreement with expectations is further verification that the field data is reliable.

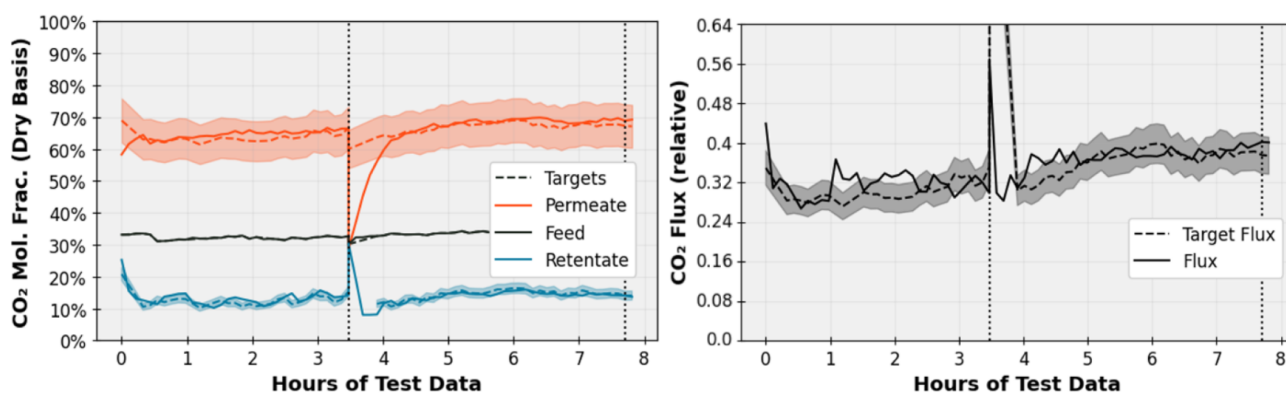


Figure 3—Comparison of pilot unit results to expectations. The plot on the left shows feed, permeate, and retentate CO₂ purities chronologically in comparison to target bands for the same operating conditions. The plot on the right shows that the CO₂ flux is also meeting expectations. These results support the reliability of the field results.

After data reliability was confirmed by comparison with expected targets for purity and flux results, trends in flux and stream purities were investigated by operating the membrane at a range of commercially relevant conditions. These operating condition variation tests were incorporated into the first phase of testing.

Phase 1.1 - Effect of Feed Pressure. Feed pressure drives flux and product purities by manipulating the driving force for permeation. As the feed pressure increases, the driving force increases. Increased driving force results in higher purity of CO₂ in the permeate stream and lower purity CO₂ in the retentate stream. This higher driving force also increases the CO₂ flux. The higher flux and product purities with increased pressure occur at the expense of higher energy requirements for compression. The increase in flux and purities is shown in Figure 4.

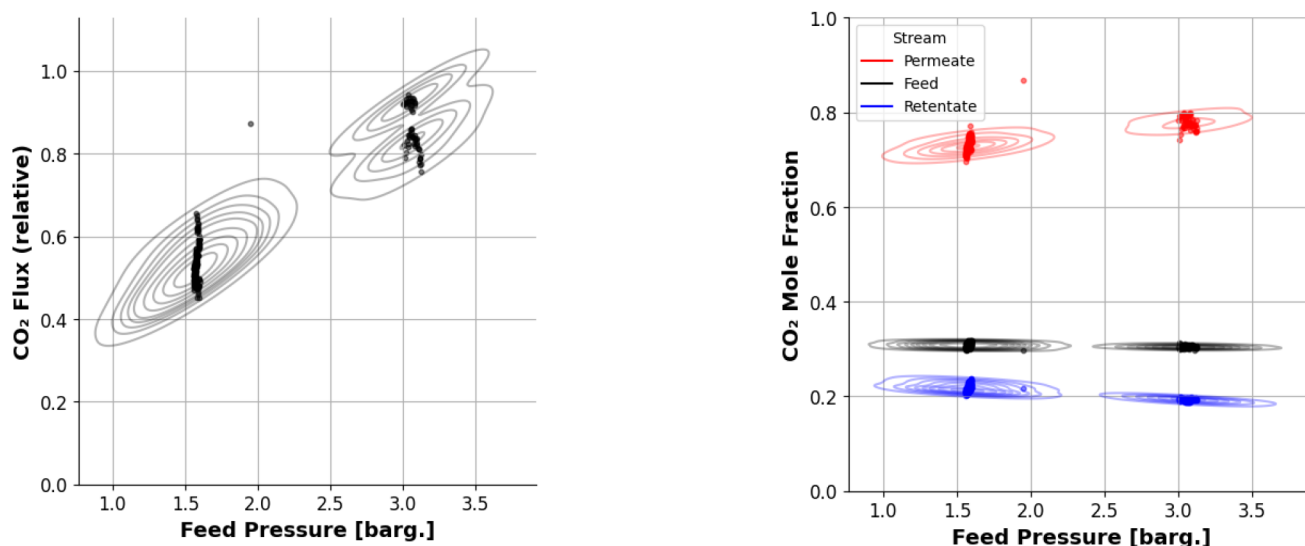


Figure 4—Condition variation results from operation condition variation experiments showing increased CO₂ flux and enrichment / depletion of CO₂ with increased feed pressure.

Phase 1.2 - Effect of Feed Flowrate and Stage Cut. By manipulating the feed flowrate, the stage cut was varied across a commercially relevant window of 20% to 40%. This range is optimal as it represents a balanced approach that is efficient for commercial-scale capture. A membrane may show excellent performance at an extremely low stage cut, but that data is not relevant for a real-world, multi-stage commercial system that must achieve high recovery rates without excessive recirculation. In commercial systems designs, the stage cut determines the appropriate membrane area for targeting product purities.

Figure 5 shows the flux and purity results across the stage cut range tested. On one hand, as the stage cut increases, CO_2 purity in the permeate decreases due to less CO_2 available for permeation on the feed side. On the other hand, the increased CO_2 recovery at increased stage cut results from a greater depletion of CO_2 in the retentate stream. The design specification of products determines the appropriate stage cut required, and in this way, the membrane area required for the separation.

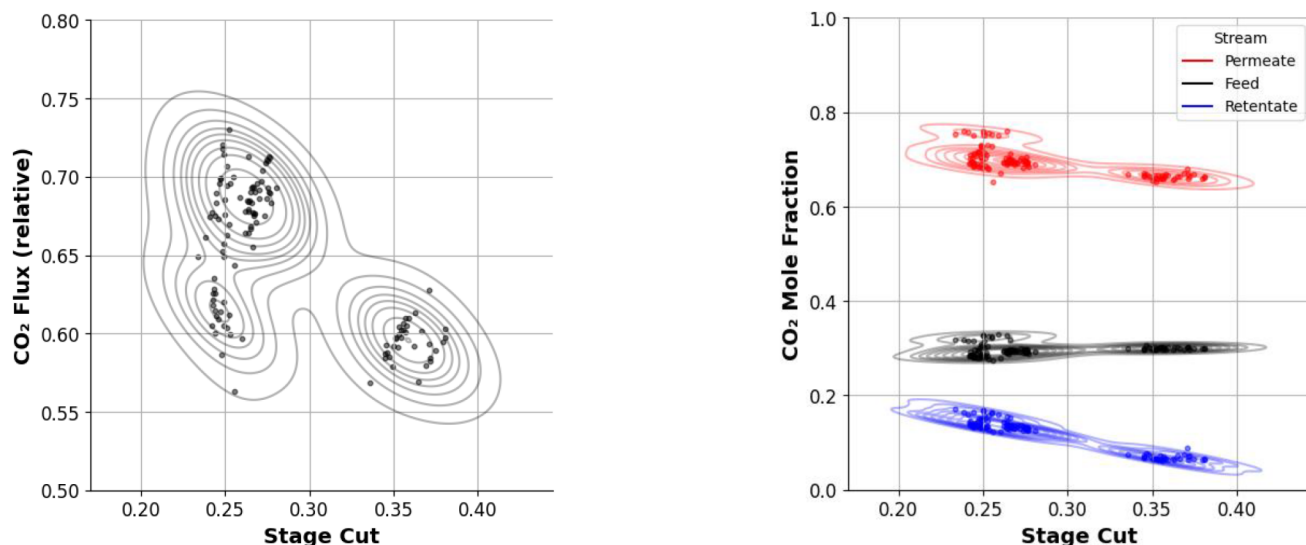


Figure 5—Trends in flux and purity with respect to stage cut manipulated by changing the feed flowrate. As the stage cut increases, the depletion of CO_2 in the retentate increases and the CO_2 purity and flux decrease. The stage cut is important for appropriate membrane sizing to target product purities.

Phase 1.3 - Effect of Humidification. Humidification is required to enhance the permeation of the FTM. Figure 6 shows results from a test to demonstrate the gain in performance when transitioning from an underhumidified to an appropriately humidified state. When the appropriate humidification is introduced during the red band, the CO_2 flux increases dramatically while the purity of leaving streams increases due to the improved membrane properties and increase in stage cut. Adequate humidification is an important design consideration for these FTM carbon capture systems.

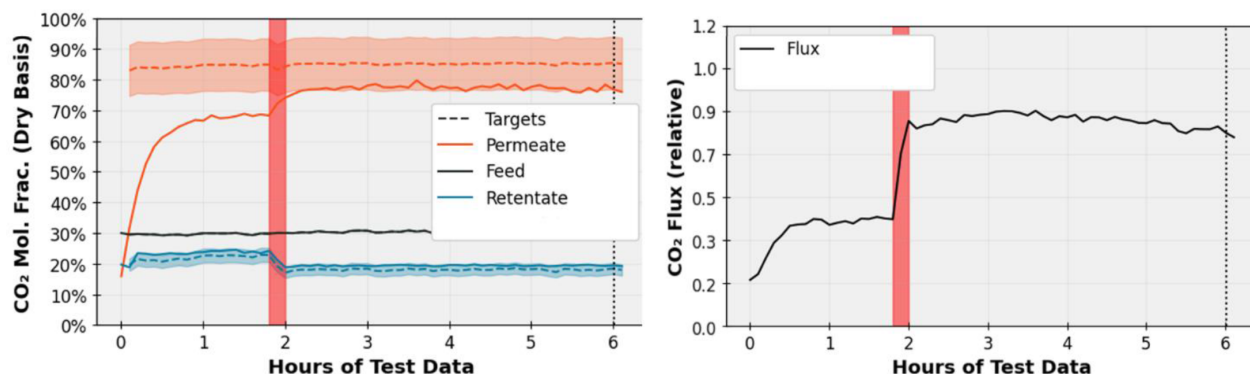


Figure 6—Flux and purity increase when transitioning from an underhumidified to an appropriately humidified state. Appropriate humidification is introduced during the red band.

The first phase of testing verified function of the pilot system via comparison of pilot unit results to the expected targets. After verifying functionality, the tunability of flux and purity by manipulating feed pressure and stage cut was performed to demonstrate the flexibility of the membrane system for targeting

purities for design purposes. The importance of the humidification system for enhancing permeation rates and product purities was also explored in this testing phase.

Testing Phase 2: Stability while treating flue gas

After completing the condition variation experiments, a longer-term test was conducted to monitor changes in membrane flux and purity over time at a commercially relevant operating condition. Throughout the test, the membrane's flux and CO₂ enrichment remained generally stable. Small variations in flux and enrichment throughout the test are caused by drifting feed flow rates and pressures within a band. The membrane pilot unit is not equipped with any system to remove acid gases or their oxidative degradation products from the membrane feed stream. Therefore, the pilot unit feed contains all the contaminants present in the flue stream. The stability test shows that the acid gases and their oxidative degradation products present in the feed did not cause a meaningful change in membrane performance.

The panel of graphs shown in Figure 7 demonstrates stability of the flux and CO₂ purities throughout the test. In this figure, vertical dotted lines indicate that a shut down and start-up has occurred. The areas in gray are removed due to high mass balance error during those times caused by malfunction of the analyzer's sampling system; Figure 2 shows the periods of high mass balance error during the greyed-out bands, operating hours 0 to 35 and 320 to 400.

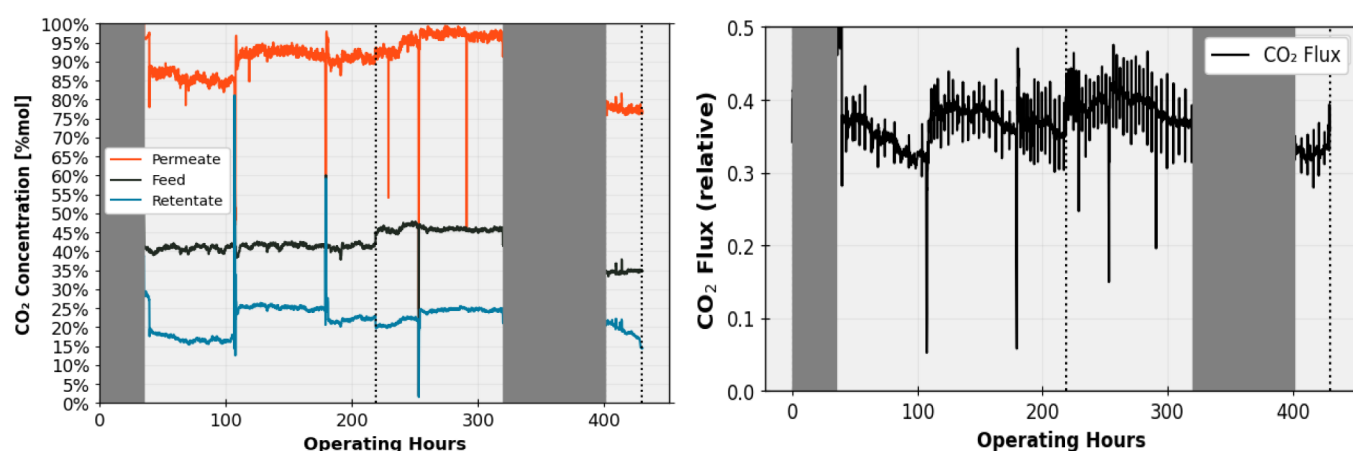


Figure 7—Panel showing results from the stability test including the CO₂ mole fractions in the feed, retentate, and permeate streams (upper plot) and CO₂ flux across the membrane (lower plot).

Operational Learnings

While the test program focused on the suitability of the membrane technology for carbon capture of untreated flue gas streams, there were a number of operational learnings that we can benefit from. 90% of downtime in the field was the result of constraints and operational challenges unrelated to the membrane unit. By identifying and addressing these challenges, we have developed a clear roadmap for improving the robustness and efficiency of our larger-scale designs.

The pie chart in Figure 8 illustrates the major causes of operational downtime we encountered, with the largest contributors being the humidification system, heat exchanger corrosion, and the CO₂ analyzer. The following section details these key learnings and the resulting design improvements we're implementing in future systems.

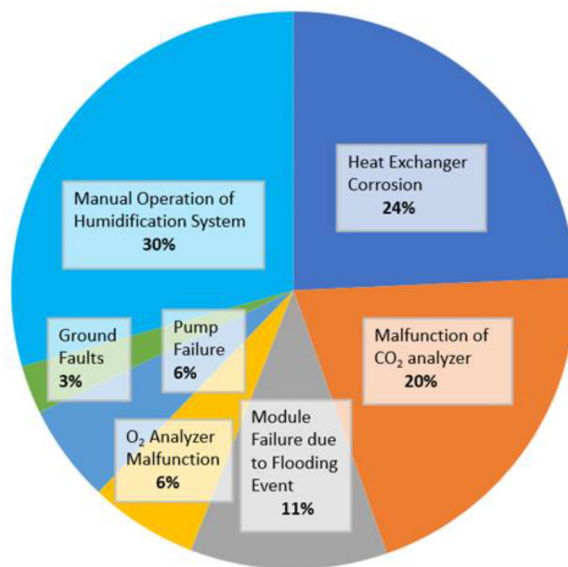


Figure 8—Summary of causes of downtime.

Some of these learnings were already incorporated into similar units operating at other sites as well as in larger units being operated at two US Oil & Gas majors, resulting in a significant improvement in ease of operation and pilot uptime.

Membranes. The pilot demonstration operated for 981 hours between May 6th 2024 and March 27th 2025. Throughout this time, three membranes were tested, with operating hours during test campaigns on each membrane, including startups and shutdowns, shown in Table 2. Module 1 and 2 were operated during the day for short tests at varied operating conditions and shutdown overnight. Module 2 experienced 23 shutdown/startup cycles before being removed to install a new membrane for the stability test. These cycles of operation and downtime demonstrate the membrane's robustness to startups and shutdowns.

Table 2—Operating Times for membranes tested at the Pilot site.

Membrane	Operating Hours	Startups / Shutdowns
Module 1	335	10
Module 2	106	23
Module 3	430	7

Over the course of the testing, one membrane failure occurred following a process upset where the feed side of the membrane housing was flooded with liquid water multiple times. Figure 9 shows time series data before and after the failure of Module 1. Two pressurizations of water that had flooded the feed (high-pressure) side of this module during the shutdown from June 17th 2024 to July 25th 2024 caused an apparent increase in N₂ permeation rate, which is reflected here by the increase in nitrogen flux after flooding. Permeate CO₂ purity becomes off-spec following the flooding event. The increase in N₂ permeation causes a decrease in CO₂ enrichment and depletion in leaving streams.

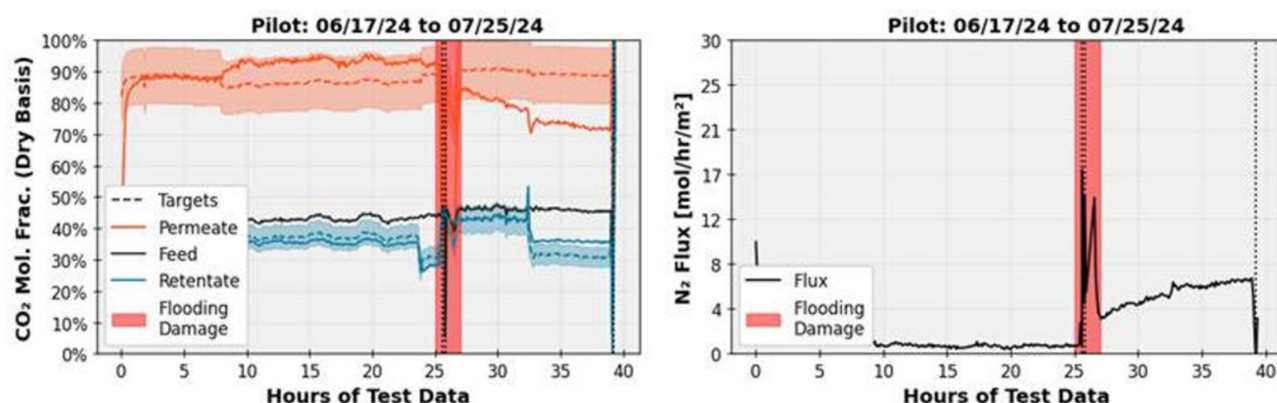


Figure 9—Time series plots showing the failure of Module 1; particularly the increase in N₂ flux and loss of separation efficiency following flooding damage.

While this is not a concern for commercial operation or for larger pilot deployments, where automated valve control would prevent such flooding from happening, it was a possible issue with these units that have some hand-operated valves. The operating methods for this pilot unit have been updated to mitigate further risks of liquid water being fed directly to the high pressure side of the module.

Other Operation Learnings. As previously stated, 90% of downtime in the field was the result of constraints and operational challenges unrelated to the membrane unit. We detail below each of these issues and how they are being addressed in future deployments.

This pilot unit did not utilize automated flow and pressure control; a commercial system would certainly use automated control valves for flow and pressure control. This resolves the drifting in operating points during long term runs for larger scale system designs.

Figure 8 notes that 30% of downtime, the largest downtime cause, occurred due to the manual operation of the humidification system. The humidification system relied on recycling condensate to provide humidification to the membrane. This originally required manual operator intervention and led to unit shutdowns due to operator availability. The unit was subsequently modified in order to automatically pump water between condensate vessels. This change was implemented before the stability testing campaign and allowed for continuous operation of the system during the stability test. Further improvements, including automated draining of vessels, have been implemented for the redeployment of this unit at another site to further extend the pilot unit's unattended runtime. Learnings from this pilot's humidification management system will be transferred to larger systems for ease of operation of the humidification system.

Figure 8 also notes that 24% of downtime was caused by corrosion of a heat exchanger. The corroded heat exchanger functions as the feed compressor aftercooler. This heat exchanger failed twice due to corrosion caused by acid gases in the feed stream. Investigation into the cause of heat exchanger corrosion suggested that several factors in combination provided the conditions necessary for a failure to occur. Table 3 below summarizes those factors and how they can be mitigated in the development of larger demonstration and commercial systems:

Table 3—Heat Exchanger Corrosion: Factors and Mitigations

Factor	Mitigation
Weld workmanship	Enforcement of NDE inspection and test requirements from API 660
Pooling of condensate inside heat exchanger	Adjusted orientation and/or inclusion of a drain line to prevent pooling
Wall thickness and chemical resistance	Increased corrosion allowance for concentrated acids and/or use of a high-nickel alloy (330 or similar)

For this pilot, the process gas was ultimately re-routed to bypass the heat exchanger, since it was determined in this case that compressor discharge gas did not require cooling. Changes to pilot unit design, including vessel coatings, are considered options for retrofitting to minimize corrosion to process equipment for small scale pilots.

The third largest cause for downtime was malfunction of the pilot unit's CO₂ and O₂ analyzer. While analyzer failure does not directly affect the operation of the pilot unit, analysis of feed, permeate, and retentate streams is required for interpreting pilot unit data. As testing progressed, increased spare inventory for analyzer replacement parts reduced the response time to address analyzer failures. The most frequent cause of analyzer malfunction was the failure of the onboard analyzer pump. The analyzer solution employed in the pilot system was selected and implemented based on lead time and ease of integration. For larger demonstration and commercial systems, it is planned to utilize TDLAS analyzers with a track record of field reliability in a wide range of industrial processes. Additionally, provisions will be added to dry analyzer sample streams and thereby prevent condensation from damaging equipment or influencing instrument readings.

Conclusion

Field testing of the facilitated transport membrane carbon capture technology at the sour gas station Aderklaa confirmed that this technology offers a very strong opportunity to deliver a low cost and low energy solution for decarbonization in refining use cases. Economics for a full-scale commercial system calculated using the performance of the membranes in the field show that levelized cost of capture of \$28-43/ton CO₂ and energy of capture of 154 kWh/ton can be achieved, which is well within the current ETS price of € 71-72/ton CO₂.

Additionally, the membranes have shown exceptional resistance to contaminants in the flue gas stream. They were able to provide CO₂ enrichment of 40 mol.% - 55 mol.% CO₂ with a single membrane stage over 430 hours of operation with high levels of acid gases and their oxidative degradation products present in the feed. Finally, the membranes have demonstrated durability to pressurization cycles, operating for tens of startup and shutdown cycles.

Future Outlook

The current field unit has demonstrated the performance of this facilitated transport membrane technology at small scale, as well as its resistance to field mechanical, thermal and chemical conditions, and in particular the high SO_x content in the flue gas. A number of improvements for larger systems that have been identified during the field testing campaign are being actively incorporated into design, increasing unit uptime and moving toward full scale deployment.

The next major proof point will be to validate the performance of the technology with commercial-scale membranes in a multi-stage system demonstrating the end-to-end process, from flue gas to liquefied CO₂, proving both the purity and recovery targets for the application. Assuming an initial commercial launch for carbon capture from a sour gas incinerator flue, with CO₂ content on the order of 10 ktpa, it is anticipated that the next deployment will utilize a multi-stage system targeting a scale of 5% of full commercial size.

References

- International Energy Agency. Emissions from Oil and Gas Operations in Net Zero Transitions A World Energy Outlook Special Report on the Oil and Gas Industry and COP28. Paris: IEA, 2023.
- International Energy Agency. *The role of CCUS in low-carbon power systems*. Paris: IEA, 2020.
- Joseph D. Henry Jr., Michael E. Prudich, William Eykamp, Ph.D., T. Alan Hatton, Keith P. Johnston, Richard M. Lemert, Robert Lemlich, Charles G. Moyers, John Newman, Herbert A. Pohl, Kent Pollock, Michael P. Thien. "Alternative Separation Processes." In *Perry's Chemical Engineers' Handbook; Seventh Edition*, by Robert H. Perry, 22-37–22-67. New York: McGraw-Hill, 1997.

- Robert F Dejaco, Brandon Burghardt, Christine Parrish. "Modeling, Simulation, and Pilot Prediction of a Facilitated-Transport Membrane with Two-phase Permeate Flow for Olefin-Paraffin Separation." *Industrial & Engineering Chemistry Research*, 2025: 3947.
- Runhong Qi, Michael A. Henson. "Approximate modeling of spiral-wound gas permeators." *journal of MEMBRANE SCIENCE*, **1996**: 11–24.
- Shah, Minish M. "Engineering Design of a Linde-BASF Amine Technology for H2 Plant Flue Gas." 2022 *Carbon Management Project Review Meeting*. Pittsburgh, PA: Linde Inc., U.S. Department of Energy, 2022. 1–16.
- Kaushik, M. (2024, May 21). Technology outlook: Carbon capture. Lux Research. <https://luxresearchinc.com/resources/utilities/technology-outlook-carbon-capture/>